

Assignment 3

HTML Iframes, Forms & Head Elements — Solutions

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Q1. Embed a YouTube video using an iframe (percentage width/height)

File: q1_youtube_iframe.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q1: Embedding a YouTube video using an iframe with percentage-based sizing -->
  <meta charset="UTF-8">
  <title>Q1 - YouTube Video Embed</title>
</head>
<body>
  <h1>My Favorite YouTube Video</h1>

  <!-- Width and height are given in percentage values so the video
  resizes automatically when the browser window is resized -->
  <iframe
    src="https://www.youtube.com/embed/dQw4w9WgXcQ"
    width="80%"
    height="60%"
    title="YouTube video player"
    allowfullscreen>
  </iframe>
</body>
</html>
```

Q2. Embed a Google Map using an iframe with title, width, height, loading="lazy"

File: q2_google_map.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q2: Embedding a Google Map using an iframe -->
  <meta charset="UTF-8">
  <title>Q2 - Google Map Embed</title>
</head>
<body>
  <h1>Location of My City</h1>

  <!-- The loading="lazy" attribute tells the browser to delay loading
  the iframe's content until it is about to become visible in the
  viewport. This improves initial page load speed and saves
  bandwidth if the user never scrolls down to the map. -->
  <iframe
    src="https://www.google.com/maps/embed?pb=!1m18!1m12!1m3!1d3556.9!2d77.6737!3d27.4924!2m3!1f0!2f0!3f0!3m2!1i1024!2i768!4f13.1"
    title="Map showing Mathura, Uttar Pradesh"
    width="600"
    height="450"
    loading="lazy">
  </iframe>
</body>
</html>
```

Q3. Two iframes stacked: introduction page and hobby page

File: q3_two_iframes.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q3: Two iframes displayed one below the other -->
  <meta charset="UTF-8">
  <title>Q3 - Two Stacked Iframes</title>
</head>
<body>
  <h1>Introduction and Hobby Pages</h1>

  <!-- First iframe: my introduction page -->
  <iframe
    src="q3_intro.html"
    title="Introduction Page"
    width="500"
    height="200">
  </iframe>

  <!-- Second iframe: my favorite hobby page -->
  <iframe
    src="q3_hobby.html"
    title="Hobby Page"
    width="500"
    height="200">
  </iframe>
</body>
</html>
```

File: q3_intro.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Content page loaded inside the first iframe of Q3 -->
  <meta charset="UTF-8">
  <title>Introduction</title>
</head>
<body>
  <h2>About Me</h2>
  <p>Hello! My name is Kunal Kumar Singh. I am a student who enjoys learning
    web development and exploring new technologies.</p>
</body>
</html>
```

File: q3_hobby.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Content page loaded inside the second iframe of Q3 -->
  <meta charset="UTF-8">
  <title>My Hobby</title>
</head>
<body>
  <h2>My Favorite Hobby</h2>
  <p>My favorite hobby is playing cricket. I enjoy playing with my friends
    during weekends and watching international matches.</p>
</body>
</html>
```

Q4. Frameset layout: navigation on top, welcome message on bottom

File: q4_frameset.html

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Frameset//EN" "http://www.w3.org/TR/html4/frameset.dtd">
<html lang="en">
<head>
  <!-- Q4: Frameset layout with a navigation frame on top and a
    welcome message frame on the bottom -->
  <title>Q4 - Frameset Layout</title>
</head>

<!-- The frameset splits the browser window into two rows:
    20% for the navigation links and 80% for the welcome message -->
<frameset rows="20%,80%">
  <frame src="q4_nav.html" name="topFrame" title="Navigation Frame">
  <frame src="q4_welcome.html" name="bottomFrame" title="Welcome Frame">
</frameset>

</html>
```

File: q4_nav.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Top frame content: navigation links for Q4 -->
  <meta charset="UTF-8">
  <title>Navigation</title>
</head>
<body>
  <a href="#">Home</a> |
  <a href="#">About</a> |
  <a href="#">Contact</a>
</body>
</html>
```

File: q4_welcome.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Bottom frame content: welcome message for Q4 -->
  <meta charset="UTF-8">
  <title>Welcome</title>
</head>
<body>
  <h2>Welcome to My Website</h2>
</body>
</html>
```

Q5. Complete head section (charset, author, viewport, title) + multilingual paragraph

File: q5_head_utf8.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q5: Complete head section demonstrating UTF-8 encoding -->

  <!-- Declares the character encoding so special characters display correctly -->
  <meta charset="UTF-8">

  <!-- Identifies the author of the document -->
  <meta name="author" content="Kunal Kumar Singh">

  <!-- Makes the page responsive on mobile devices -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Q5 - UTF-8 Encoding Test</title>
</head>
<body>
  <h1>UTF-8 Multilingual Text Test</h1>

  <p>
    English: Hello, welcome to my website!<br>
    Hindi: नमस्ते, मेरी वेबसाइट पर आपका स्वागत है!<br>
    Japanese: こんにちは、私のウェブサイトへようこそ！
  </p>
</body>
</html>
```

Q6. User Registration Form (Full Name, Email, Password, Confirm Password)

File: q6_registration_form.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q6: User Registration Form -->
  <meta charset="UTF-8">
  <title>Q6 - User Registration Form</title>
</head>
<body>
  <h1>User Registration Form</h1>

  <form action="#" method="post">
    <label for="fullname">Full Name:</label><br>
    <input type="text" id="fullname" name="fullname" required><br><br>

    <label for="email">Email Address:</label><br>
    <input type="email" id="email" name="email" required><br><br>

    <label for="password">Password:</label><br>
    <input type="password" id="password" name="password" required><br><br>

    <label for="confirm_password">Confirm Password:</label><br>
    <input type="password" id="confirm_password" name="confirm_password" required><br><br>

    <input type="submit" value="Register">
  </form>
</body>
</html>
```

Q7. Login Form with hidden CSRF token field

File: q7_login_form.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q7: Login Form with a hidden CSRF token -->
  <meta charset="UTF-8">
  <title>Q7 - Login Form</title>
</head>
<body>
  <h1>Login Form</h1>

  <form action="#" method="post">
    <label for="username">Username:</label><br>
    <input type="text" id="username" name="username" required><br><br>

    <label for="password">Password:</label><br>
    <input type="password" id="password" name="password" required><br><br>

    <!-- A CSRF (Cross-Site Request Forgery) token is a unique, secret
         value generated by the server and embedded in the form. When
         the form is submitted, the server checks that this token
         matches the one it issued. This proves the request genuinely
         came from the site's own form and not from a malicious
         external site tricking the user's browser into submitting it. -->
    <input type="hidden" name="csrf_token" value="alb2c3d4e5f6dummy">

    <input type="submit" value="Login">
  </form>
</body>
</html>

```

Q8. Feedback Form (textarea, radio buttons, submit)

File: q8_feedback_form.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q8: Feedback Form with textarea and radio buttons -->
  <meta charset="UTF-8">
  <title>Q8 - Feedback Form</title>
</head>
<body>
  <h1>Feedback Form</h1>

  <form action="#" method="post">
    <label for="feedback">Your Feedback:</label><br>
    <textarea id="feedback" name="feedback" rows="5" cols="40"></textarea><br><br>

    <p>Rate our service:</p>
    <input type="radio" id="good" name="rating" value="Good">
    <label for="good">Good</label><br>

    <input type="radio" id="average" name="rating" value="Average">
    <label for="average">Average</label><br>

    <input type="radio" id="poor" name="rating" value="Poor">
    <label for="poor">Poor</label><br><br>

    <input type="submit" value="Submit Feedback">
  </form>
</body>
</html>

```

Q9. File Upload Form restricted to .jpg / .png

File: q9_file_upload_form.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q9: File Upload Form restricted to .jpg and .png files -->
  <meta charset="UTF-8">
  <title>Q9 - File Upload Form</title>
</head>
<body>
  <h1>Upload Your Photo</h1>

  <form action="#" method="post" enctype="multipart/form-data">
    <label for="photo">Choose an image file:</label><br>
    <input type="file" id="photo" name="photo" accept=".jpg,.png" required><br><br>

    <input type="submit" value="Upload">
  </form>

  <p>Note: Only image files with the .jpg or .png extension are allowed to be uploaded.</p>
</body>
</html>

```

Q10. Contact Form with placeholder and required attributes

File: q10_contact_form.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q10: Contact Form using placeholder and required attributes -->
  <meta charset="UTF-8">
  <title>Q10 - Contact Form</title>
</head>
<body>
  <h1>Contact Us</h1>

  <form action="#" method="post">
    <label for="fullname">Full Name:</label><br>
    <input type="text" id="fullname" name="fullname" placeholder="Enter your full name" required><br><br>

    <label for="email">Email:</label><br>
    <input type="email" id="email" name="email" placeholder="Enter your email address" required><br><br>

    <label for="subject">Subject:</label><br>
    <input type="text" id="subject" name="subject" placeholder="Enter the subject" required><br><br>

    <label for="message">Message:</label><br>
    <textarea id="message" name="message" rows="5" cols="40" placeholder="Type your message here" required></textarea><br><br>

    <input type="submit" value="Send Message">
  </form>
</body>
</html>
```

Q11. Student Information Form with course dropdown

File: q11_student_form.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q11: Student Information Form with a course dropdown -->
  <meta charset="UTF-8">
  <title>Q11 - Student Information Form</title>
</head>
<body>
  <h1>Student Information Form</h1>

  <form action="#" method="post">
    <label for="name">Name:</label><br>
    <input type="text" id="name" name="name" required><br><br>

    <label for="roll">Roll Number:</label><br>
    <input type="text" id="roll" name="roll" required><br><br>

    <label for="course">Course:</label><br>
    <select id="course" name="course" required>
      <option value="" disabled selected>Select Course</option>
      <option value="bca">BCA</option>
      <option value="bsc_cs">B.Sc Computer Science</option>
      <option value="btech">B.Tech</option>
      <option value="mca">MCA</option>
    </select><br><br>

    <input type="submit" value="Submit">
  </form>
</body>
</html>
```

Q12. Paragraph using HTML entities

File: q12_html_entities.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q12: Demonstrating HTML entities -->
  <meta charset="UTF-8">
  <title>Q12 - HTML Entities</title>
</head>
<body>
  <h1>HTML Entities Demonstration</h1>

  <p>
    Less than: &lt;<br>
    Greater than: &gt;<br>
    Ampersand: &amp;<br>
    Double quote: &quot;<br>
    Single quote: &apos;<br>
    Copyright symbol: &copy;<br>
    Trademark symbol: &trade;<br>
    Registered symbol: &reg;
  </p>
</body>
</html>
```

Q13. Webpage with text in English, Hindi, and French

File: q13_three_languages.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q13: Page with text in English, Hindi, and French -->
  <meta charset="UTF-8">
  <title>Q13 - Multilingual Page</title>
</head>
<body>
  <h1>Text in Three Languages</h1>

  <p>English: Welcome to my website.</p>
  <p>Hindi: मेरी वेबसाइट पर आपका स्वागत है।</p>
  <p>French: Bienvenue sur mon site web.</p>
</body>
</html>
```

Q14. Embed an external website (Wikipedia) using an iframe

File: q14_external_iframe.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q14: Embedding an external website using an iframe -->
  <meta charset="UTF-8">
  <title>Q14 - External Website Embed</title>
</head>
<body>
  <h1>Embedded External Website</h1>

  <iframe
    src="https://www.wikipedia.org"
    title="Wikipedia Website"
    width="700"
    height="450">
  </iframe>

  <p>Some websites may not allow embedding for security reasons.</p>
</body>
</html>
```

Q15. Payment Form (Cardholder Name, Card Number, Expiry Date, CVV)

File: q15_payment_form.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q15: Payment Form -->
  <meta charset="UTF-8">
  <title>Q15 - Payment Form</title>
</head>
<body>
  <h1>Payment Details</h1>

  <form action="#" method="post">
    <label for="cardholder">Cardholder Name:</label><br>
    <input type="text" id="cardholder" name="cardholder" required><br><br>

    <label for="cardnumber">Card Number:</label><br>
    <input type="text" id="cardnumber" name="cardnumber" maxlength="16" required><br><br>

    <label for="expiry">Expiry Date:</label><br>
    <input type="month" id="expiry" name="expiry" required><br><br>

    <label for="cvv">CVV:</label><br>
    <input type="password" id="cvv" name="cvv" maxlength="3" required><br><br>

    <input type="submit" value="Pay Now">
  </form>
</body>
</html>

```

Q16. Demonstrate URI encoding in a hyperlink

File: q16_uri_encoding.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q16: Demonstrating URI encoding in a hyperlink -->
  <meta charset="UTF-8">
  <title>Q16 - URI Encoding</title>
</head>
<body>
  <h1>URI Encoding Demonstration</h1>

  <p>Original URL (contains a space and an ampersand, which are special characters):</p>
  <p>https://example.com/search?query=HTML & CSS</p>

  <p>Correctly encoded URL (spaces replaced with %20, ampersand encoded as %26):</p>
  <a href="https://example.com/search?query=HTML%20%26%20CSS">
    https://example.com/search?query=HTML%20%26%20CSS
  </a>
</body>
</html>

```

Q17. form.html loads result.html inside an iframe after Submit

File: q17_form.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q17: form.html - submits into an iframe using the target attribute,
       so no JavaScript is required to load result.html -->
  <meta charset="UTF-8">
  <title>Q17 - Form with Iframe Target</title>
</head>
<body>
  <h1>Sample Form</h1>

  <!-- The form's target attribute matches the iframe's name attribute,
       so clicking Submit loads result.html directly inside the iframe -->
  <form action="q17_result.html" method="get" target="resultFrame">
    <label for="text_input">Enter some text:</label>
    <input type="text" id="text_input" name="text_input">
    <input type="submit" value="Submit">
  </form>

  <iframe name="resultFrame" title="Result Frame" width="400" height="100"></iframe>
</body>
</html>

```

File: q17_result.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q17: Result page loaded inside the iframe after form submission -->
  <meta charset="UTF-8">
  <title>Result</title>
</head>
<body>
  <p>Form successfully submitted.</p>
</body>
</html>

```

Q18. Long quotation (blockquote) and short quotation (q) with an HTML entity

File: q18_quotations.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q18: Long and short quotations using blockquote and q -->
  <meta charset="UTF-8">
  <title>Q18 - Quotations</title>
</head>
<body>
  <h1>Quotations Example</h1>

  <!-- Long quotation -->
  <blockquote>
    Education is the most powerful weapon which you can use to change
    the world, and it works best when hard work &amp; dedication go
    hand in hand.
  </blockquote>

  <!-- Short quotation inside a paragraph -->
  <p>As my teacher always says, <q>practice makes a student perfect</q>, and I truly believe that.</p>
</body>
</html>

```

Q19. Form security attributes: autocomplete="off" and required

File: q19_form_security.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q19: Form Security Attributes -->
  <meta charset="UTF-8">
  <title>Q19 - Form Security Attributes</title>
</head>
<body>
  <h1>Secure Login Form</h1>

  <form action="#" method="post" autocomplete="off">
    <label for="username">Username:</label><br>
    <input type="text" id="username" name="username" required><br><br>

    <label for="password">Password:</label><br>
    <input type="password" id="password" name="password" required><br><br>

    <input type="submit" value="Login">
  </form>

  <p>Turning off autocomplete prevents the browser from storing and
  auto-filling sensitive information such as passwords. This reduces
  the risk of that data being exposed if the device is shared,
  stolen, or accessed by someone else.</p>
</body>
</html>

```

Q20. All important head elements: title, meta, base, link, style, script

File: q20_head_elements.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Q20: Demonstrating all important head elements -->

  <!-- title: sets the text shown on the browser tab and in search results -->
  <title>Q20 - Important Head Elements</title>

  <!-- meta charset: defines the character encoding used by the document -->
  <meta charset="UTF-8">

  <!-- meta description: gives search engines a summary of the page content -->
  <meta name="description" content="A demonstration page showing important HTML head elements.">

  <!-- meta author: states who created the document -->
  <meta name="author" content="Kunal Kumar Singh">

  <!-- base: sets the default base URL for all relative links and resources on the page -->
  <base href="https://example.com/">

  <!-- link: connects an external resource, here a favicon icon, to the page -->
  <link rel="icon" href="favicon.ico" type="image/x-icon">

  <!-- style: reserved for internal CSS rules (left empty as CSS is not used here) -->
  <style>
</style>

  <!-- script: reserved for internal JavaScript code (left empty as JS is not used here) -->
  <script>
</script>
</head>
<body>
  <h1>Important Head Elements Demonstrated</h1>
  <p>View the page source or the head section above to see each element explained with comments.</p>
</body>
</html>
```